RELATIVE DIMENSION IS UPPER SEMI-CONTINUOUS, AND LOCALLY CONSTANT FOR SMOOTH MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-geometry.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for stacks-geometry.tex (ge-occupied.json, 2026-10-14); 207 route=geometry / chapter-list field hits in focused-seeds are alias noise (route name and alphabetical chapter enumerations), excluded per the ladder alias-blind-spot caveat.

Source reference. Original lemma 0DRN: sources/stacks-chow/tags.txt, source lines 17228–17228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03BX: sources/stacks-chow/tags.txt, source lines 4364–4364. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 04NT: sources/stacks-chow/tags.txt, source lines 6091–6091. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02NM: sources/stacks-chow/tags.txt, source lines 3485–3485. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02G1: sources/stacks-chow/tags.txt, source lines 3212–3212. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{T} \to \mathcal{X}$ be a locally of finite type morphism of algebraic stacks.*

1. *The function $t \mapsto \dim_t(\mathcal{T}_{f(t)})$ is upper semi-continuous on $|\mathcal{T}|$.*
2. *If f is smooth, then the function $t \mapsto \dim_t(\mathcal{T}_{f(t)})$ is locally constant on $|\mathcal{T}|$.*

Proof. Suppose to begin with that $\mathcal{T}$ is a scheme T, let $U \to \mathcal{X}$ be a smooth surjective morphism whose source is a scheme, and let $T' = T \times_{\mathcal{X}} U$. Let $f' : T' \to U$ be the pull-back of f over U, and let $g : T' \to T$ be the projection.

Lemma 0DRN shows that $\dim_{t'}(T'_{f'(t')}) = \dim_{g(t')}(T_{f(g(t'))})$, for $t' \in T'$, while, since g is smooth and surjective (being the base-change of a smooth surjective morphism) the map induced by g on underlying topological spaces is continuous and open (by Properties of Spaces, Lemma 03BX), and surjective. Thus it suffices to note that

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part (1) for the morphism f' follows from Morphisms of Spaces, Lemma 04NT, and part (2) from either of Morphisms, Lemma 02NM or Morphisms, Lemma 02G1 (each of which gives the result for schemes, from which the analogous results for algebraic spaces can be deduced exactly as in Morphisms of Spaces, Lemma 04NT.

Now return to the general case, and choose a smooth surjective morphism $h : V \to \mathcal{T}$ whose source is a scheme. If $v \in V$, then, essentially by definition, we have

$$\dim_{h(v)}(\mathcal{T}_{f(h(v))}) = \dim_v(V_{f(h(v))}) - \dim_v(V_{h(v)}).$$

Since V is a scheme, we have proved that the first of the terms on the right hand side of this equality is upper semi-continuous (and even locally constant if f is smooth), while the second term is in fact locally constant. Thus their difference is upper semi-continuous (and locally constant if f is smooth), and hence the function $\dim_{h(v)}(\mathcal{T}_{f(h(v))})$ is upper semi-continuous on $|V|$ (and locally constant if f is smooth). Since the morphism $|V| \to |\mathcal{T}|$ is open and surjective, the lemma follows. $\square$